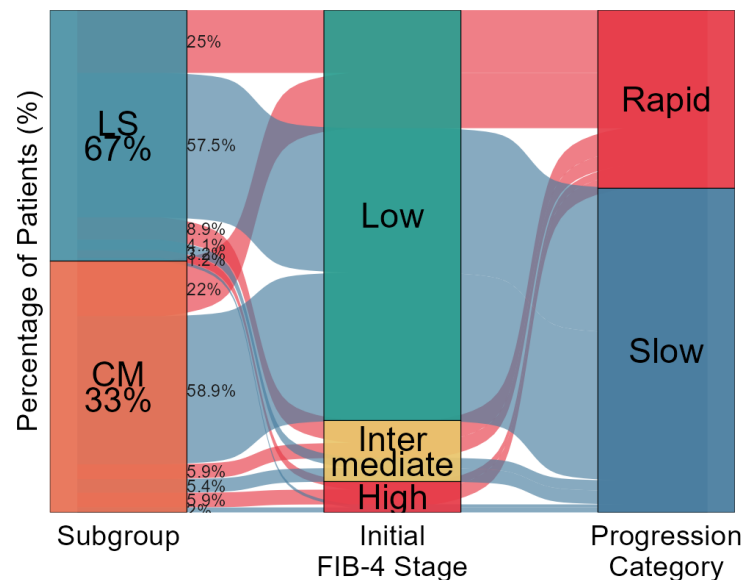


pRFPS: A Subgroup-Specific Interpretable Risk Score for Predicting Rapid Fibrosis Progression in Metabolic Dysfunction-Associated Steatotic Liver Disease

Supplemental information	Description
Supplemental Fig. 1	Alluvial plot of MASLD phenotypes, initial FIB-4 risk, and progression categories in Tapestry Study.
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Continuous pRFPS construction: As a sensitivity analysis, we constructed a continuous pRFPS using the same subgroup-specific predictors and SHAP-derived weights. Predictor values were min–max normalized to a 0–1 range using the MCB training-set minimum and maximum. Predictors inversely associated with rapid progression were reverse scaled so that higher values consistently indicated higher risk. The continuous pRFPS was calculated as the weighted sum of these risk-aligned normalized predictor values. This analysis assessed whether retaining continuous feature values improved performance relative to the threshold-based pRFPS.

Conflict of interests: A.J.A. reports grants or contracts from Rhythm Pharmaceuticals, Boehringer Ingelheim, Vivus Inc., Vivus Pharmaceuticals, Regeneron, and Novo Nordisk, all paid to the institution. A.J.A. also reports royalties or licenses from Phenomix Sciences; personal consulting fees from Boehringer Ingelheim, Regeneron, Currax, and Structure Pharmaceutical; and consulting fees (paid to the institute) from Amgen, RareStone, and Bausch Health. Additional disclosures include personal honoraria from Eli Lilly and Boehringer Ingelheim, honoraria (paid to the institute) from Vivus Pharmaceuticals for educational and professional activities, and institutional travel support from Currax. A.J.A. holds 10 patents, including 3 issued patents; has served on Data Safety Monitoring Boards or Advisory Boards for Amgen, Boehringer Ingelheim, Structure Pharmaceutical, Currax, and Regeneron; and holds stock options in Gila Therapeutics and Phenomix Sciences. All disclosures relate to activities within the past 36 months. The remaining authors declare no competing interests related to this work.



Supplemental Fig. 1: Alluvial plot of MASLD phenotypes, initial FIB-4 risk, and progression categories in Tapestry Study.

Supplemental Table 1. ICD code definitions for clinical comorbidities and liver-related outcomes.

Disease Name	ICD-10 / ICD-9 Codes
MASLD	K76.0, K75.8, K74.60, K74.69, 571.8, 571.9, 571.5,
Hypertension	I10–I16
Dyslipidemia	E78.x
Type 2 diabetes mellitus	E11.x
Chronic kidney disease	N18.x, N19
Cardiovascular disease	I20–I27, I38, I40–I51
Depression	F32.x, F33.x, F34.1, F41.2, F43.2
Fibrosis	K74.0
Advanced fibrosis / Cirrhosis	K74.60, K74.69, 571.5
Hepatocellular carcinoma (HCC)	C22.x
Liver transplantation	T86.4, Z94.4, 996.8, V42.7

Supplemental Table 2. Cohort inclusion criteria for fibrosis progression analysis

Inclusion criteria	MCB cohort (n)	Tapestry cohort (n)
Initial cohort	5,747	7,480
≥1 cardiometabolic risk factor	5,520	7,158
After excluding patients with missing longitudinal FIB-4 measurements	3,019	4,041
After excluding patients with additional missing clinical variables	2,422	3,093
After excluding patients with <5 years of follow-up	752	1,240

Supplemental Table 3. Distribution of missing clinical variables in the Mayo Clinic Biobank (MCB) and Tapestry cohorts

Variable	MCB (n = 5,520)	Tapestry (n = 7,158)
	Missing n (%)	Missing n (%)
Age (years)	98 (1.8%)	327 (4.6%)
Sex	39 (0.7%)	257 (3.6%)
BMI (kg/m ²)	49 (0.9%)	257 (3.6%)
HbA1c (%)	1,523 (27.6%)	2,276 (31.8%)
ALT (U/L)	714 (12.9%)	898 (12.5%)
AST (U/L)	587 (10.6%)	817 (11.4%)
ALP (U/L)	712 (12.9%)	895 (12.5%)
Platelet Count (×10 ⁹ /L)	1,740 (31.5%)	4,178 (58.4%)
Albumin (g/dL)	691 (12.5%)	883 (12.3%)
Triglycerides (mg/dL)	985 (17.8%)	1,653 (23.1%)
HDL (mg/dL)	887 (16.1%)	1,581 (22.1%)
LDL (mg/dL)	1,036 (18.8%)	1,723 (24.1%)
BMI = body mass index; ALT = alanine aminotransferase; AST = aspartate aminotransferase; ALP = alkaline phosphatase; LDL = low-density lipoprotein cholesterol; HDL = high-density lipoprotein cholesterol;		

Supplemental Table 4: Comprehensive model hyperparameter search space for machine learning model development

Category	Model	Hyperparameters
Linear	Logistic Regression	C ∈ {0.01, 0.1, 1, 10}; penalty = L2; solver = lbfgs; max_iter = 3000; class_weight ∈ {None, balanced}
Instance	kNN	n_neighbors ∈ {3, 5, 7}; weights ∈ {uniform, distance}
Kernel	SVM	C ∈ {0.1, 1}; kernel = RBF; gamma = scale; class_weight ∈ {None, balanced}; probability = True
Tree	Decision Tree	max_depth ∈ {3, 5, 7}; min_samples_leaf ∈ {2, 5}; class_weight ∈ {None, balanced}
Ensemble	Random Forest	n_estimators ∈ {50, 100}; max_depth ∈ {5, 10, None}; min_samples_split ∈ {2, 5}; class_weight ∈ {None, balanced}
Boosting	Gradient Boosting	n_estimators ∈ {50, 100}; max_depth ∈ {3, 5, 7}; learning_rate ∈ {0.01, 0.1}
	XGBoost	n_estimators ∈ {50, 100}; max_depth ∈ {3, 5, 7}; learning_rate ∈ {0.01, 0.1}; subsample ∈ {0.8, 1.0}; scale_pos_weight ∈ {1, 3, 5}
	LightGBM	n_estimators ∈ {50, 100}; max_depth ∈ {3, 5, 7}; learning_rate ∈ {0.01, 0.1}; num_leaves ∈ {15, 31}; class_weight ∈ {None, balanced}
Neural Network	MLP	hidden_layer_sizes ∈ {(50,), (100,)}; learning_rate_init ∈ {0.001, 0.01}; alpha ∈ {0.0001, 0.001}; solver = adam; early_stopping = True; validation_fraction = 0.1; max_iter = 500
LR = Logistic Regression; kNN = k-Nearest Neighbors; SVM = Support Vector Machine; DT = Decision Tree; RF = Random Forest; GB = Gradient Boosting; XGB = eXtreme Gradient Boosting; LGBM = Light Gradient Boosting Machine; MLP = Multi-Layer Perceptron.		

Supplemental Table 5. Latent class analysis fit statistics in MCB dataset

Number of classes	n	Bayesian information criterion (BIC)	Entropy	Mean posterior probability	Min posterior probability	Mean posterior Probability per class	Minimum class proportion
1	2422	25569.11	NaN	1	1	1	100
2	2422	25366.04	0.749	0.91	0.50	0.9 0.92	41.41
3	2422	25329.23	0.509	0.76	0.37	0.73 0.77 0.76	25.14
4	2422	25382.79	0.501	0.72	0.34	0.74 0.74 0.59 0.76	11.60
5	2422	25456.73	0.541	0.71	0.34	0.88 0.84 0.75 0.61 0.86	4.58
6	2422	25524.7	0.554	0.70	0.25	0.74 0.69 0.6 0.77 0.66 0.81	5.70
7	2422	25607.76	0.519	0.63	0.32	0.77 0.82 0.64 0.54 0.6 0.7 0.91	5.53
8	2422	25697.8	0.549	0.65	0.27	0.72 0.7 0.57 0.63 0.72 0.62 0.65 0.62	0.99
9	2422	25796.87	0.536	0.64	0.32	0.63 0.57 0.59 0.55 0.5 0.63 0.85 0.64 0.75	0.95
10	2422	25869.49	0.669	0.71	0.34	0.74 0.74 0.62 0.62 0.63 0.64 0.75 0.63 0.77 0.8	1.03

Supplemental Table 6. Class-specific distribution of maximum posterior probabilities for the selected two-class solution.

Maximum posterior probability	LS, n (%)	CM, n (%)
0.50–<0.60	98 (9.8)	118 (8.3)
0.60–<0.70	74 (7.4)	140 (9.9)
0.70–<0.80	30 (3.0)	108 (7.6)
0.80–<0.90	26 (2.6)	84 (5.9)
≥0.90	775 (77.3)	969 (68.3)

Supplemental Table 7. MASLD phenotypes characteristics in MCB cohort.

Characteristic	LS (n=1,419)	CM (n=1,003)	Overall (n=2,422)	P-value
Demographics				
Female	806 (56.8%)	520 (51.8%)	1326 (54.7%)	0.0163
Age (years)	59.7 (12.9)	61.5 (12.2)	60.5 (12.7)	0.00291
BMI (kg/m ²)	31.3 (6.11)	34.5 (7.15)	32.6 (6.75)	<0.001
Clinical Lab Variables				
HbA1c (%)	5.51 (0.444)	7.18 (0.981)	6.20 (1.09)	<0.001
ALT (U/L)	40.7 (54.3)	40.6 (36.8)	40.6 (47.8)	0.0526
AST (U/L)	38.5 (40.6)	39.9 (44.4)	39.1 (42.2)	0.144
ALP (U/L)	92.4 (45.6)	100 (67.4)	95.7 (55.8)	<0.001
Albumin (g/dL)	7.61 (25.4)	22.6 (82.6)	13.8 (57.2)	<0.001
BUN (mg/dL)	18.2 (7.23)	20.9 (9.07)	19.3 (8.15)	<0.001
LDL (mg/dL)	93.3 (32.2)	80.5 (31.8)	88.0 (32.6)	<0.001
HDL (mg/dL)	61.4 (49.7)	58.5 (84.8)	60.2 (66.5)	<0.001
Triglycerides (mg/dL)	128 (59.8)	189 (98.9)	153 (83.8)	<0.001
Platelet count (×10 ⁹ /L)	224 (84.9)	227 (93.3)	225 (88.5)	0.793
AST/ALT ratio	1.06 (0.391)	1.04 (0.358)	1.05 (0.378)	0.342
AST/Platelet ratio	0.226 (0.367)	0.255 (0.602)	0.238 (0.479)	0.436
NFS	-3.09 (16.8)	-11.9 (54.5)	-6.76 (37.7)	<0.001
Liver Disease & Genetics				
MASH	332 (23.4%)	252 (25.1%)	584 (24.1%)	0.335
PRS-3	1.21 (0.976)	1.25 (1.02)	1.23 (0.994)	0.591
PRS-5 (High)	251 (17.7%)	157 (15.7%)	408 (16.8%)	0.32

PNPLA3 1/1	134 (9.4%)	110 (11.0%)	244 (10.1%)	0.054
TM6SF2 0/1+1/1	239 (16.8%)	191 (19.0%)	430 (17.8%)	0.247
HSD17B13 1/1	91 (6.4%)	65 (6.5%)	156 (6.4%)	0.293
Comorbidities				
Heart disease	1,013 (71.4%)	774 (77.2%)	1,787 (73.8%)	0.001
CKD	541 (38.1%)	545 (54.3%)	1,086 (44.8%)	<0.001
Depression	638 (45.0%)	467 (46.6%)	1,105 (45.6%)	0.456
Hypertension	1,133 (79.8%)	909 (90.6%)	2,042 (84.3%)	<0.001
Dyslipidemia	1,162 (81.9%)	901 (89.8%)	2,063 (85.2%)	<0.001
Diabetes	398 (28.0%)	931 (92.8%)	1,329 (54.9%)	<0.001
CMR count	2.63 (0.980)	3.97 (0.846)	3.19 (1.14)	<0.001
Continuous variables are presented as mean (SD); categorical variables as n (%). BMI = body mass index; ALT = alanine aminotransferase; AST = aspartate aminotransferase; ALP = alkaline phosphatase; BUN = blood urea nitrogen; BUN = blood urea nitrogen; LDL = low-density lipoprotein cholesterol; HDL = high-density lipoprotein cholesterol; NFS = NAFLD fibrosis score; CMR = cardiometabolic risk; CKD = chronic kidney disease; MASH = metabolic-associated steatohepatitis. P values were calculated using the Wilcoxon rank-sum test for continuous variables and chi-square or Fisher's exact tests for categorical variables, as appropriate. Bold text indicates variables used for phenotyping.				

Supplemental Table 8. MASLD phenotypes characteristics in Tapestry cohort.

Characteristic	LS (n=2,099)	CM (n=994)	Overall (n=3,093)	P-value
Demographics				
Female	1,303 (62.1%)	534 (53.7%)	1,837 (59.4%)	<0.001
Age (years)	54.1 (13.1)	56.0 (11.9)	54.7 (12.7)	<0.001
BMI (kg/m²)	32.1 (5.68)	34.3 (5.89)	32.8 (5.84)	<0.001
Clinical Lab Variables				
HbA1c (%)	5.47 (0.426)	6.91 (0.942)	5.93 (0.928)	<0.001
ALT (U/L)	43.7 (37.5)	50.3 (59.2)	45.8 (45.7)	<0.001
AST (U/L)	39.2 (31.8)	43.0 (37.1)	40.4 (33.6)	<0.001
ALP (U/L)	93.7 (57.6)	95.2 (42.9)	94.2 (53.3)	0.002
Albumin (g/dL)	7.35 (29.6)	11.6 (44.5)	8.73 (35.1)	<0.001
BUN	16.4 (7.04)	17.9 (7.25)	16.9 (7.14)	<0.001
LDL (mg/dL)	101 (36.8)	85.5 (30.0)	95.9 (35.5)	<0.001
HDL (mg/dL)	70.6 (188)	68.0 (63.5)	69.8 (159)	0.001
Triglycerides (mg/dL)	134 (63.7)	203 (136)	156 (98.6)	<0.001
Platelet count (($\times 10^9$ /L)	238 (81.0)	237 (95.4)	238 (85.9)	0.19
AST/ALT ratio	0.981 (0.399)	0.942 (0.348)	0.969 (0.384)	0.002
AST/Platelet ratio	0.222 (0.492)	0.236 (0.346)	0.227 (0.450)	<0.001
NFS	-3.32 (19.5)	-5.21 (29.3)	-3.93 (23.2)	<0.001
Liver Disease & Genetics				
MASH	538 (25.6%)	294 (29.6%)	832 (26.9%)	0.021
PRS-5 (High)	387 (18.4%)	162 (16.3%)	549 (17.7%)	0.202
PRS-3	0.816 (1.01)	0.770 (0.999)	0.801 (1.01)	0.194
PNPLA3 1/1	165 (7.9%)	82 (8.2%)	247 (8.0%)	0.188
TM6SF2 0/1+1/1	246 (11.7%)	95 (9.6%)	341 (11.0%)	0.038
HSD17B13 1/1	71 (3.4%)	46 (4.6%)	117 (3.8%)	0.121
Comorbidities				
Heart disease	1,224 (58.3%)	656 (66.0%)	1,880 (60.8%)	<0.001
CKD	480 (22.9%)	368 (37.0%)	848 (27.4%)	<0.001
Depression	985 (46.9%)	464 (46.7%)	1,449 (46.8%)	0.908
Hypertension	1,450 (69.1%)	859 (86.4%)	2,309 (74.7%)	<0.001
Dyslipidemia	1,581 (75.3%)	878 (88.3%)	2,459 (79.5%)	<0.001
Diabetes	589 (28.1%)	857 (86.2%)	1,446 (46.8%)	<0.001
CMR count	2.51 (0.978)	3.86 (0.808)	2.94 (1.12)	<0.001

Continuous variables are presented as mean (SD); categorical variables as n (%). BMI = body mass index; ALT = alanine aminotransferase; AST = aspartate aminotransferase; ALP = alkaline phosphatase; BUN = blood urea nitrogen; BUN = blood urea nitrogen; LDL = low-density lipoprotein cholesterol; HDL = high-density lipoprotein cholesterol; NFS = NAFLD fibrosis score; CMR = cardiometabolic risk; CKD = chronic kidney disease; MASH = metabolic-associated steatohepatitis. P values were calculated using the Wilcoxon rank-sum test for continuous variables and chi-square or Fisher's exact tests for categorical variables, as appropriate. Bold text indicates variables used for phenotyping.

Supplemental Table 9. Patient characteristics according to subgroup-specific progression category in MCB

Characteristic	LS				CM			
	Overall	Rapid	Slow	P- value	Overall	Rapid	Slow	P- value
	(n=427)	(n=160)	(n=267)		(n=325)	(n=126)	(n=199)	
Demographics								
Age (years)	58.9 (13.9)	57.3 (12.6)	59.8 (14.5)	0.021	61.8 (12.1)	60.3 (10.8)	62.8 (12.8)	0.024
Female	239 (56%)	73 (46%)	166 (62%)	0.001	164 (50%)	63 (50%)	101 (51%)	>0.9
BMI (kg/m²)	31.3 (6.4)	30.7 (6.4)	31.7 (6.4)	0.089	34.5 (7.7)	33.4 (6.4)	35.2 (8.4)	0.1
Metabolic Profile								
HbA1c (%)	5.5 (0.5)	5.4 (0.5)	5.5 (0.4)	0.11	7.2 (0.9)	7.2 (0.9)	7.2 (1.0)	0.3
ALT (U/L)	42.5 (38.1)	47.0 (41.3)	39.8 (35.9)	0.023	48.7 (54.3)	51.9 (46.6)	46.6 (58.6)	0.023
AST (U/L)	44.1 (41.2)	54.3 (57.1)	38.1 (25.8)	<0.001	51.7 (70.2)	62.8 (91.7)	44.6 (51.2)	<0.001
ALP (U/L)	102.0 (53.8)	115.6 (65.1)	93.8 (44.0)	<0.001	112.7 (74.7)	128.7 (96.4)	102.4 (54.5)	<0.001
Platelet Count (×10⁹/L)	214.1 (94.4)	179.6 (88.9)	234.8 (91.7)	<0.001	216.4 (112.5)	162.1 (89.2)	250.7 (112.3)	<0.001
Albumin (g/dL)	7.6 (26.9)	6.2 (13.2)	8.4 (32.5)	0.12	15.6 (42.4)	8.7 (18.7)	19.9 (51.7)	<0.001
Triglycerides (mg/dL)	126.2 (63.9)	121.6 (64.9)	129.0 (63.2)	0.03	179.1 (84.5)	169.6 (95.5)	185.1 (76.4)	0.019
HDL (mg/dL)	63.2 (58.3)	62.3 (85.3)	63.7 (33.1)	0.004	59.9 (77.6)	54.4 (25.5)	63.3 (97.0)	>0.9
LDL (mg/dL)	90.2 (30.1)	87.9 (27.9)	91.6 (31.3)	0.2	78.2 (26.6)	78.6 (23.1)	77.9 (28.6)	0.5
BUN (mg/dL)	18.5 (8.2)	21.3 (10.3)	16.8 (6.0)	<0.001	21.1 (9.5)	22.7 (10.0)	20.2 (9.0)	0.021
NFS	-2.8 (17.9)	-1.4 (8.7)	-3.7 (21.5)	<0.001	-7.1 (28.1)	-1.9 (12.7)	-10.4 (34.0)	<0.001
Liver Disease & Genetics								
MASH	122 (29%)	60 (38%)	62 (23%)	0.002	128 (39%)	77 (61%)	51 (26%)	<0.001
HCC	37 (8.7%)	34 (21%)	3 (1.1%)	<0.001	30 (9.2%)	22 (17%)	8 (4.0%)	<0.001
Transplant	73 (17%)	65 (41%)	8 (3.0%)	<0.001	52 (16%)	45 (36%)	7 (3.5%)	<0.001
PRS-3				0.4				>0.9

1	104 (24%)	32 (20%)	72 (27%)		82 (25%)	33 (26%)	49 (25%)	
2	163 (38%)	58 (36%)	105 (39%)		115 (35%)	41 (33%)	74 (37%)	
3	111 (26%)	49 (31%)	62 (23%)		79 (24%)	30 (24%)	49 (25%)	
4	43 (10%)	18 (11%)	25 (9.4%)		37 (11%)	17 (13%)	20 (10%)	
5	4 (0.9%)	2 (1.3%)	2 (0.7%)		10 (3.1%)	4 (3.2%)	6 (3.0%)	
6	2 (0.5%)	1 (0.6%)	1 (0.4%)		2 (0.6%)	1 (0.8%)	1 (0.5%)	
PRS-5	0.2 (0.3)	0.2 (0.4)	0.1 (0.3)	0.003	0.1 (0.3)	0.2 (0.3)	0.1 (0.3)	0.3
PNPLA3 G/G	49 (11%)	33 (21%)	16 (6.0%)	<0.001	47 (14%)	23 (18%)	24 (12%)	0.3
TM6SF2 T/T	5 (1.2%)	3 (1.9%)	2 (0.7%)	0.5	4 (1.2%)	1 (0.8%)	3 (1.5%)	0.7
HSD17B13 non-A/non-A	22 (5.2%)	5 (3.1%)	17 (6.4%)	0.3	26 (8.0%)	8 (6.3%)	18 (9.0%)	0.6
Comorbidities								
Diabetes	147 (34%)	67 (42%)	80 (30%)	0.016	297 (91%)	120 (95%)	177 (89%)	0.077
Hypertension	366 (86%)	137 (86%)	229 (86%)	>0.9	300 (92%)	116 (92%)	184 (92%)	>0.9
Dyslipidemia	341 (80%)	117 (73%)	224 (84%)	0.01	298 (92%)	112 (89%)	186 (93%)	0.2
Depression	208 (49%)	63 (39%)	145 (54%)	0.004	167 (51%)	64 (51%)	103 (52%)	>0.9
Heart Disease	320 (75%)	116 (73%)	204 (76%)	0.4	259 (80%)	104 (83%)	155 (78%)	0.4
Kidney Disease	200 (47%)	81 (51%)	119 (45%)	0.3	175 (54%)	79 (63%)	96 (48%)	0.015
Follow-up visits, n	10.8 (11.4)	11.3 (13.0)	10.4 (10.3)	0.046	10.7 (11.0)	12.8 (14.9)	9.4 (7.4)	0.8
Continuous variables are presented as mean (SD); categorical variables as n (%). BMI = body mass index; ALT = alanine aminotransferase; AST = aspartate aminotransferase; ALP = alkaline phosphatase; BUN = blood urea nitrogen; BUN = blood urea nitrogen; LDL = low-density lipoprotein cholesterol; HDL = high-density lipoprotein cholesterol; NFS = NAFLD fibrosis score; CMR = cardiometabolic risk; CKD = chronic kidney disease; MASH = metabolic-associated steatohepatitis. P values were calculated using the Wilcoxon rank-sum test for continuous variables and chi-square or Fisher's exact tests for categorical variables, as appropriate. Bold text indicates variables used for phenotyping.								

Supplemental Table 10. Patient characteristics according to subgroup-specific progression category in Tapestry Study.

Characteristic	LS				CM			
	Overall	Rapid	Slow	P- value	Overall	Rapid	Slow	P- value
	(n=831)	(n=309)	(n=522)		(n=409)	(n=138)	(n=271)	
Demographics								
Age (years)	53.8 (13.6)	56.7 (11.6)	52.1 (14.3)	<0.001	55.7 (12.2)	57.1 (10.3)	54.9 (13.0)	0.2
Female	517 (62%)	146 (47%)	371 (71%)	<0.001	240 (59%)	76 (55%)	164 (61%)	0.3

BMI (kg/m ²)	32.0 (5.8)	30.7 (5.8)	32.8 (5.6)	<0.001	34.2 (5.9)	33.0 (6.4)	34.7 (5.6)	0.008
Metabolic Profile								
HbA1c (%)	5.4 (0.5)	5.4 (0.5)	5.5 (0.4)	<0.001	7.0 (1.0)	7.1 (0.9)	6.9 (1.0)	0.006
MASH	253 (30%)	148 (48%)	105 (20%)	<0.001	160 (39%)	94 (68%)	66 (24%)	<0.001
ALT (U/L)	48.1 (41.5)	57.4 (48.0)	42.5 (36.1)	<0.001	55.0 (83.1)	56.7 (39.1)	54.1 (98.2)	<0.001
AST (U/L)	45.1 (39.0)	60.4 (48.7)	36.1 (28.3)	<0.001	46.5 (42.0)	60.1 (51.6)	39.6 (34.2)	<0.001
ALP (U/L)	100.4 (55.2)	119.8 (76.2)	88.8 (32.5)	<0.001	101.4 (43.2)	120.7 (55.9)	91.7 (30.9)	<0.001
Platelet Count (×10 ⁹ /L)	230.5 (91.6)	179.5 (86.2)	260.6 (80.8)	<0.001	235.3 (118.7)	178.7 (88.4)	264.1 (121.8)	<0.001
Albumin (g/dL)	8.3 (33.5)	7.4 (38.8)	8.8 (29.9)	<0.001	10.8 (57.5)	8.0 (17.5)	12.2 (69.5)	0.14
Triglycerides (mg/dL)	133.0 (61.4)	123.0 (54.0)	138.9 (64.8)	<0.001	199.8 (148.6)	189.8 (135.4)	204.8 (154.9)	0.3
HDL (mg/dL)	80.3 (296.9)	58.3 (21.3)	93.3 (373.8)	<0.001	66.3 (44.4)	57.7 (28.8)	70.6 (50.0)	0.002
LDL (mg/dL)	99.4 (37.3)	90.5 (29.6)	104.6 (40.3)	<0.001	84.5 (30.5)	85.9 (34.4)	83.8 (28.3)	0.8
BUN (mg/dL)	16.9 (7.9)	20.0 (9.9)	15.1 (5.6)	<0.001	18.0 (7.3)	20.5 (9.2)	16.8 (5.7)	<0.001
NFS	-3.8 (22.2)	-2.3 (25.7)	-4.6 (19.7)	<0.001	-4.6 (37.9)	-1.8 (11.8)	-6.0 (45.8)	<0.001
Liver Disease & Genetics								
MASH	253 (30%)	148 (48%)	105 (20%)	<0.001	160 (39%)	94 (68%)	66 (24%)	<0.001
HCC	55 (6.6%)	48 (16%)	7 (1.3%)	<0.001	27 (6.6%)	24 (17%)	3 (1.1%)	<0.001
Transplant	121 (15%)	108 (35%)	13 (2.5%)	<0.001	53 (13%)	47 (34%)	6 (2.2%)	<0.001
PRS-3				0.9				0.016
1	434 (52%)	167 (54%)	267 (51%)		210 (51%)	57 (41%)	153 (56%)	
2	181 (22%)	63 (20%)	118 (23%)		95 (23%)	36 (26%)	59 (22%)	
3	142 (17%)	52 (17%)	90 (17%)		73 (18%)	28 (20%)	45 (17%)	
4	60 (7.2%)	21 (6.8%)	39 (7.5%)		24 (5.9%)	14 (10%)	10 (3.7%)	
5	14 (1.7%)	6 (1.9%)	8 (1.5%)		6 (1.5%)	2 (1.4%)	4 (1.5%)	
6	-	-	-		1 (0.2%)	1 (0.7%)	0 (0%)	
PRS-5	0.2 (0.3)	0.2 (0.3)	0.1 (0.3)	<0.001	0.1 (0.3)	0.2 (0.3)	0.1 (0.3)	0.064
PNPLA3 G/G	73 (8.8%)	36 (12%)	37 (7.1%)	0.08	43 (11%)	26 (19%)	17 (6.3%)	<0.001
TM6SF2 T/T	5 (0.6%)	1 (0.3%)	4 (0.8%)	0.6	1 (0.2%)	0 (0%)	1 (0.4%)	0.4
HSD17B13 non-A/non-A	27 (3.2%)	6 (1.9%)	21 (4.0%)	0.035	21 (5.1%)	6 (4.3%)	15 (5.5%)	0.2

Comorbidities								
Diabetes	266 (32%)	119 (39%)	147 (28%)	0.003	357 (87%)	129 (93%)	228 (84%)	0.012
Hypertension	600 (72%)	235 (76%)	365 (70%)	0.068	354 (87%)	120 (87%)	234 (86%)	>0.9
Dyslipidemia	612 (74%)	199 (64%)	413 (79%)	<0.001	357 (87%)	109 (79%)	248 (92%)	<0.001
Depression	416 (50%)	130 (42%)	286 (55%)	<0.001	194 (47%)	57 (41%)	137 (51%)	0.1
Heart Disease	502 (60%)	197 (64%)	305 (58%)	0.15	266 (65%)	92 (67%)	174 (64%)	0.7
Kidney Disease	215 (26%)	124 (40%)	91 (17%)	<0.001	163 (40%)	68 (49%)	95 (35%)	0.008
Follow-up visits, n	11.8 (13.0)	14.6 (16.8)	10.1 (9.7)	0.7	12.0 (14.4)	14.6 (20.7)	10.7 (9.7)	0.5
Continuous variables are presented as mean (SD); categorical variables as n (%). BMI = body mass index; ALT = alanine aminotransferase; AST = aspartate aminotransferase; ALP = alkaline phosphatase; BUN = blood urea nitrogen; BUN = blood urea nitrogen; LDL = low-density lipoprotein cholesterol; HDL = high-density lipoprotein cholesterol; NFS = NAFLD fibrosis score; CMR = cardiometabolic risk; CKD = chronic kidney disease; MASH = metabolic-associated steatohepatitis. P values were calculated using the Wilcoxon rank-sum test for continuous variables and chi-square or Fisher's exact tests for categorical variables, as appropriate. Bold text indicates variables used for phenotyping.								

Supplemental Table 11: Performance of the ML models for predicting rapid fibrosis progression in MCB test set.

Setting	Model	Accuracy	F1	AUROC	AUPRC	Brier	Precision	Recall
Overall	LR	0.628 [0.550-0.702]	0.622 [0.542-0.697]	0.704 [0.618-0.787]	0.609 [0.474-0.724]	0.223 [0.196-0.252]	0.514 [0.397-0.632]	0.668 [0.544-0.780]
Overall	KNN	0.634 [0.556-0.709]	0.433 [0.373-0.500]	0.693 [0.606-0.777]	0.621 [0.493-0.731]	0.228 [0.188-0.269]	0.946 [0.000-1.000]	0.051 [0.000-0.113]
Overall	RF	0.681 [0.609-0.749]	0.651 [0.575-0.727]	0.711 [0.621-0.787]	0.612 [0.477-0.725]	0.211 [0.178-0.244]	0.601 [0.456-0.739]	0.515 [0.388-0.643]
Overall	DT	0.616 [0.543-0.689]	0.595 [0.518-0.672]	0.604 [0.505-0.696]	0.477 [0.366-0.593]	0.283 [0.231-0.340]	0.502 [0.377-0.631]	0.517 [0.393-0.643]
Overall	GB	0.715 [0.642-0.782]	0.700 [0.625-0.771]	0.749 [0.663-0.822]	0.671 [0.540-0.780]	0.200 [0.165-0.240]	0.625 [0.492-0.754]	0.654 [0.527-0.773]
Overall	SVM	0.676 [0.602-0.742]	0.664 [0.585-0.734]	0.703 [0.612-0.783]	0.592 [0.458-0.715]	0.210 [0.179-0.243]	0.570 [0.448-0.689]	0.651 [0.526-0.765]
Overall	MLP	0.648 [0.570-0.722]	0.640 [0.560-0.714]	0.707 [0.623-0.789]	0.623 [0.500-0.731]	0.208 [0.181-0.237]	0.535 [0.419-0.648]	0.668 [0.547-0.789]
Overall	XGB	0.689 [0.616-0.755]	0.633 [0.551-0.710]	0.706 [0.619-0.786]	0.625 [0.493-0.745]	0.265 [0.213-0.319]	0.660 [0.500-0.812]	0.397 [0.258-0.521]
Overall	LGBM	0.647 [0.576-0.715]	0.633 [0.562-0.705]	0.712 [0.627-0.794]	0.639 [0.515-0.747]	0.224 [0.182-0.266]	0.538 [0.415-0.661]	0.599 [0.476-0.719]
LS	LR	0.663 [0.558-0.756]	0.659 [0.558-0.756]	0.788 [0.694-0.875]	0.690 [0.527-0.833]	0.211 [0.192-0.230]	0.530 [0.388-0.667]	0.811 [0.666-0.935]
LS	KNN	0.571 [0.465-0.674]	0.561 [0.460-0.663]	0.713 [0.612-0.818]	0.529 [0.379-0.678]	0.210 [0.171-0.249]	0.462 [0.344-0.590]	0.937 [0.844-1.000]
LS	RF	0.741 [0.640-0.826]	0.720 [0.622-0.811]	0.809 [0.718-0.894]	0.647 [0.481-0.832]	0.178 [0.148-0.214]	0.651 [0.486-0.812]	0.653 [0.480-0.815]
LS	DT	0.615 [0.500-0.721]	0.611 [0.500-0.717]	0.728 [0.610-0.834]	0.603 [0.426-0.763]	0.235 [0.172-0.307]	0.488 [0.341-0.635]	0.751 [0.600-0.889]
LS	GB	0.720 [0.616-0.803]	0.704 [0.599-0.799]	0.777 [0.674-0.865]	0.601 [0.444-0.762]	0.193 [0.157-0.234]	0.608 [0.438-0.763]	0.688 [0.513-0.839]
LS	SVM	0.719 [0.616-0.802]	0.704 [0.601-0.798]	0.762 [0.667-0.851]	0.600 [0.434-0.764]	0.192 [0.163-0.221]	0.608 [0.455-0.757]	0.687 [0.517-0.833]
LS	MLP	0.642 [0.547-0.744]	0.631 [0.536-0.731]	0.744 [0.637-0.840]	0.636 [0.461-0.794]	0.200 [0.156-0.243]	0.513 [0.366-0.659]	0.659 [0.484-0.829]

LS	XGB	0.685 [0.581-0.779]	0.653 [0.549-0.751]	0.768 [0.664-0.854]	0.621 [0.462-0.784]	0.202 [0.155-0.257]	0.582 [0.400-0.759]	0.530 [0.357-0.697]
LS	LGBM	0.707 [0.616-0.791]	0.681 [0.577-0.778]	0.759 [0.656-0.847]	0.567 [0.416-0.736]	0.209 [0.160-0.266]	0.608 [0.448-0.767]	0.591 [0.417-0.769]
CM	LR	0.769 [0.662-0.862]	0.758 [0.642-0.857]	0.850 [0.727-0.948]	0.850 [0.723-0.943]	0.201 [0.183-0.219]	0.675 [0.500-0.829]	0.762 [0.591-0.920]
CM	KNN	0.692 [0.585-0.800]	0.685 [0.568-0.797]	0.811 [0.685-0.914]	0.726 [0.531-0.877]	0.168 [0.122-0.219]	0.574 [0.406-0.742]	0.762 [0.588-0.913]
CM	RF	0.804 [0.708-0.892]	0.762 [0.639-0.868]	0.799 [0.667-0.914]	0.803 [0.659-0.905]	0.165 [0.123-0.213]	0.931 [0.750-1.000]	0.526 [0.321-0.714]
CM	DT	0.681 [0.569-0.785]	0.635 [0.512-0.750]	0.727 [0.588-0.856]	0.681 [0.505-0.825]	0.214 [0.153-0.285]	0.616 [0.389-0.842]	0.445 [0.259-0.636]
CM	GB	0.708 [0.585-0.815]	0.698 [0.578-0.802]	0.724 [0.565-0.853]	0.716 [0.538-0.851]	0.200 [0.165-0.235]	0.599 [0.429-0.769]	0.720 [0.535-0.893]
CM	SVM	0.799 [0.708-0.892]	0.787 [0.681-0.891]	0.842 [0.716-0.942]	0.846 [0.718-0.942]	0.156 [0.117-0.200]	0.727 [0.538-0.893]	0.760 [0.588-0.913]
CM	MLP	0.845 [0.754-0.923]	0.834 [0.740-0.923]	0.844 [0.721-0.949]	0.858 [0.732-0.947]	0.146 [0.102-0.195]	0.795 [0.625-0.950]	0.802 [0.640-0.950]
CM	XGB	0.726 [0.615-0.831]	0.663 [0.539-0.783]	0.764 [0.626-0.882]	0.760 [0.595-0.877]	0.200 [0.130-0.274]	0.772 [0.529-1.000]	0.402 [0.217-0.593]
CM	LGBM	0.632 [0.523-0.738]	0.628 [0.519-0.738]	0.754 [0.630-0.867]	0.742 [0.583-0.870]	0.212 [0.189-0.236]	0.513 [0.359-0.667]	0.760 [0.591-0.917]

LR = Logistic Regression; kNN = k-Nearest Neighbors; SVM = Support Vector Machine; DT = Decision Tree; RF = Random Forest; GB = Gradient Boosting; XGB = eXtreme Gradient Boosting; LGBM = Light Gradient Boosting Machine; MLP = Multi-Layer Perceptron.

Supplemental Table 12: Performance of the ML models for predicting rapid fibrosis progression in Tapestry Study validation set.

Setting	Model	Accuracy	F1	AUROC	AUPRC	Brier	Precision	Recall
Overall	LR	0.723 [0.697-0.750]	0.706 [0.679-0.734]	0.782 [0.757-0.812]	0.711 [0.670-0.752]	0.188 [0.179-0.195]	0.606 [0.561-0.648]	0.668 [0.625-0.712]
Overall	KNN	0.648 [0.617-0.672]	0.427 [0.407-0.446]	0.775 [0.747-0.799]	0.653 [0.603-0.697]	0.183 [0.172-0.194]	0.775 [0.615-0.939]	0.037 [0.021-0.054]
Overall	RF	0.764 [0.742-0.788]	0.724 [0.698-0.752]	0.818 [0.795-0.841]	0.727 [0.686-0.765]	0.163 [0.153-0.173]	0.741 [0.694-0.786]	0.533 [0.489-0.584]
Overall	DT	0.638 [0.609-0.662]	0.390 [0.379-0.398]	0.500 [0.500-0.500]	0.362 [0.338-0.391]	0.233 [0.224-0.243]	0.000 [0.000-0.000]	0.000 [0.000-0.000]
Overall	GB	0.756 [0.732-0.777]	0.737 [0.713-0.760]	0.812 [0.786-0.836]	0.727 [0.683-0.767]	0.163 [0.152-0.175]	0.656 [0.609-0.698]	0.682 [0.640-0.728]
Overall	SVM	0.765 [0.740-0.788]	0.740 [0.714-0.767]	0.809 [0.783-0.834]	0.739 [0.696-0.777]	0.164 [0.155-0.173]	0.693 [0.642-0.737]	0.629 [0.583-0.672]
Overall	MLP	0.754 [0.731-0.779]	0.733 [0.708-0.760]	0.798 [0.772-0.826]	0.726 [0.683-0.771]	0.173 [0.163-0.181]	0.662 [0.616-0.704]	0.654 [0.608-0.701]
Overall	XGB	0.739 [0.716-0.763]	0.669 [0.641-0.696]	0.797 [0.771-0.824]	0.706 [0.664-0.747]	0.196 [0.181-0.212]	0.782 [0.727-0.834]	0.387 [0.346-0.428]
Overall	LGBM	0.751 [0.727-0.774]	0.733 [0.706-0.758]	0.816 [0.794-0.840]	0.722 [0.681-0.763]	0.165 [0.154-0.177]	0.647 [0.605-0.690]	0.686 [0.644-0.728]
LS	LR	0.707 [0.676-0.739]	0.697 [0.666-0.729]	0.790 [0.754-0.820]	0.723 [0.666-0.773]	0.204 [0.198-0.210]	0.590 [0.545-0.646]	0.709 [0.662-0.760]
LS	KNN	0.584 [0.550-0.621]	0.583 [0.549-0.619]	0.751 [0.717-0.785]	0.624 [0.570-0.673]	0.190 [0.176-0.206]	0.469 [0.427-0.512]	0.856 [0.814-0.896]
LS	RF	0.749 [0.716-0.774]	0.721 [0.687-0.750]	0.808 [0.774-0.837]	0.720 [0.664-0.772]	0.172 [0.161-0.185]	0.699 [0.647-0.756]	0.580 [0.529-0.631]
LS	DT	0.374 [0.342-0.407]	0.272 [0.255-0.289]	0.346 [0.314-0.377]	0.320 [0.289-0.351]	0.262 [0.258-0.265]	0.374 [0.342-0.407]	1.000 [1.000-1.000]
LS	GB	0.736 [0.702-0.766]	0.717 [0.681-0.748]	0.794 [0.758-0.826]	0.703 [0.645-0.751]	0.176 [0.162-0.192]	0.652 [0.602-0.698]	0.634 [0.583-0.694]
LS	SVM	0.755 [0.727-0.787]	0.727 [0.694-0.759]	0.787 [0.751-0.818]	0.722 [0.672-0.769]	0.179 [0.167-0.192]	0.712 [0.654-0.765]	0.580 [0.530-0.635]

LS	MLP	0.732 [0.704-0.763]	0.715 [0.685-0.747]	0.775 [0.739-0.809]	0.690 [0.631-0.752]	0.180 [0.165-0.197]	0.636 [0.586-0.695]	0.660 [0.610-0.714]
LS	XGB	0.724 [0.695-0.755]	0.685 [0.651-0.718]	0.783 [0.748-0.815]	0.674 [0.614-0.727]	0.190 [0.171-0.209]	0.680 [0.613-0.737]	0.497 [0.444-0.550]
LS	LGBM	0.734 [0.704-0.765]	0.700 [0.666-0.734]	0.781 [0.745-0.812]	0.696 [0.644-0.746]	0.186 [0.168-0.205]	0.683 [0.621-0.732]	0.538 [0.480-0.591]
CM	LR	0.759 [0.717-0.794]	0.735 [0.689-0.774]	0.796 [0.752-0.840]	0.698 [0.621-0.779]	0.213 [0.206-0.220]	0.622 [0.546-0.688]	0.698 [0.624-0.762]
CM	KNN	0.690 [0.645-0.736]	0.672 [0.624-0.720]	0.738 [0.685-0.789]	0.584 [0.491-0.660]	0.187 [0.166-0.209]	0.526 [0.454-0.599]	0.691 [0.606-0.770]
CM	RF	0.747 [0.703-0.787]	0.685 [0.636-0.729]	0.778 [0.727-0.823]	0.650 [0.561-0.726]	0.174 [0.157-0.195]	0.679 [0.586-0.771]	0.457 [0.372-0.534]
CM	DT	0.333 [0.286-0.377]	0.252 [0.224-0.280]	0.498 [0.489-0.505]	0.332 [0.284-0.376]	0.539 [0.504-0.575]	0.332 [0.284-0.376]	0.992 [0.976-1.000]
CM	GB	0.663 [0.621-0.709]	0.636 [0.591-0.686]	0.705 [0.653-0.758]	0.599 [0.510-0.679]	0.194 [0.181-0.208]	0.494 [0.404-0.576]	0.596 [0.512-0.673]
CM	SVM	0.766 [0.728-0.810]	0.724 [0.679-0.773]	0.792 [0.743-0.836]	0.712 [0.643-0.779]	0.166 [0.148-0.186]	0.678 [0.590-0.761]	0.566 [0.485-0.648]
CM	MLP	0.775 [0.732-0.815]	0.739 [0.694-0.785]	0.798 [0.748-0.839]	0.711 [0.632-0.785]	0.165 [0.147-0.187]	0.682 [0.595-0.755]	0.610 [0.531-0.686]
CM	XGB	0.750 [0.707-0.792]	0.681 [0.630-0.730]	0.748 [0.695-0.798]	0.648 [0.563-0.729]	0.216 [0.185-0.249]	0.700 [0.596-0.800]	0.433 [0.342-0.508]
CM	LGBM	0.650 [0.610-0.691]	0.627 [0.581-0.671]	0.728 [0.682-0.772]	0.556 [0.464-0.649]	0.213 [0.204-0.222]	0.480 [0.397-0.551]	0.610 [0.526-0.683]

LR = Logistic Regression; kNN = k-Nearest Neighbors; SVM = Support Vector Machine; DT = Decision Tree; RF = Random Forest; GB = Gradient Boosting; XGB = eXtreme Gradient Boosting; LGBM = Light Gradient Boosting Machine; MLP = Multi-Layer Perceptron.

Supplemental Table 13: Associations of pRFPS risk group and baseline FIB-4 risk group with liver-related outcomes.

Cohort	Subgroup	Comparison	HCC OR (95% CI)	Liver transplantation OR (95% CI)
MCB	LS	High vs. low pRFPS	1.95 (0.99–3.85)	2.87 (1.70–4.83)
MCB	CM	High vs. low pRFPS	4.84 (2.17–10.76)	3.52 (1.91–6.48)
MCB	LS	High vs. low FIB-4	0.97 (0.45–2.07)	0.46 (0.24–0.89)
MCB	CM	High vs. low FIB-4	1.48 (0.67–3.24)	0.17 (0.06–0.48)
Tapestry	LS	High vs. low pRFPS	3.27 (1.87–5.73)	6.10 (4.00–9.29)
Tapestry	CM	High vs. low pRFPS	4.99 (2.18–11.45)	8.45 (4.38–16.29)
Tapestry	LS	High vs. low FIB-4	1.20 (0.60–2.38)	0.63 (0.36–1.12)
Tapestry	CM	High vs. low FIB-4	1.88 (0.79–4.46)	1.13 (0.55–2.31)

OR= Odds Ratio. LS= liver-specific; CM= cardiometabolic; HCC= hepatocellular carcinoma.

Supplementary Table 14. Model performance using a fixed 3-year rapid-progression definition.

Setting	Model	Accuracy (95% CI)	F1-macro (95% CI)	AUROC (95% CI)	Brier score (95% CI)
Overall	DT	0.657 (0.583–0.728)	0.636 (0.556–0.714)	0.711 (0.622–0.791)	0.182 (0.148–0.220)
Overall	GB	0.690 (0.609–0.755)	0.621 (0.535–0.700)	0.681 (0.583–0.778)	0.191 (0.154–0.232)
LS	RF	0.627 (0.523–0.721)	0.564 (0.456–0.672)	0.639 (0.511–0.755)	0.206 (0.169–0.243)
LS	DT	0.615 (0.512–0.721)	0.588 (0.480–0.693)	0.650 (0.527–0.767)	0.198 (0.149–0.251)
CM	LR	0.725 (0.615–0.831)	0.623 (0.492–0.751)	0.682 (0.543–0.817)	0.202 (0.141–0.268)
CM	MLP	0.712 (0.600–0.831)	0.595 (0.460–0.728)	0.680 (0.521–0.825)	0.198 (0.136–0.269)

For each approach, the model with the best performance was selected under either the fixed 3-year progression definition or the primary subgroup-specific progression thresholds. The performance of these selected models under the fixed 3-year progression definition is shown. LR = logistic regression; DT = decision tree; RF = random forest; GB = gradient boosting; MLP = multilayer perceptron; CI = Confidence Interval.

Supplementary Table 15. Performance of threshold-based and continuous pRFPS variants.

Subgroup	pRFPS variant	AUROC	F1-score
LS	Threshold-based	0.705	0.637
LS	Min-max-normalized continuous	0.633	0.576
CM	Threshold-based	0.703	0.643
CM	Min-max-normalized continuous	0.785	0.66
LS = Liver-specific; CM = Cardiometabolic.			

Supplementary Table 16. Sex- and age-stratified performance of subgroup-specific rapid fibrosis progression models in the MCB test set.

Subgroup	Stratum	Model	AUROC	F1-score
LS	All patients	RF	0.8125	0.7263
LS	Female		0.8521	0.7552
LS	Male		0.738	0.6872
LS	Age <65 years		0.7923	0.7319
LS	Age ≥65 years		0.7895	0.6711
CM	All patients	MLP	0.843	0.8375
CM	Female		0.8873	0.8194
CM	Male		0.8	0.8578
CM	Age <65 years		0.8421	0.8528
CM	Age ≥65 years		0.836	0.8168
LS = Liver-specific; CM = Cardometabolic; RF = random forest; MLP = multilayer perceptron;				